STACKS, DISCRIMINANTS: LEMMA-DUALIZING-BASE-FLAT-FLAT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 554-558 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 560-587, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 22 of queue_discriminant.json (pb=1196, stmt 554-558, proof 560-587).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00MQ: sources/stacks-discriminant/pinned-original-source.tex, source lines 587–587. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00UJ: sources/stacks-discriminant/pinned-original-source.tex, source lines 587–587. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00HT: sources/stacks-discriminant/pinned-original-source.tex, source lines 587–587. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT3: sources/stacks-discriminant/pinned-original-source.tex, source lines 450–454. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT5: sources/stacks-discriminant/pinned-original-source.tex, source lines 493–497. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a flat quasi-finite homomorphism of Noetherian rings. Then $\omega_{B/A}$ is a flat A-module.*

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $\mathfrak{q} \subset B$ be a prime lying over $\mathfrak{p} \subset A$. We will show that the localization $\omega_{B/A,\mathfrak{q}}$ is flat over $A_\mathfrak{p}$. This suffices by Algebra, Lemma 00HT. By Algebra, Lemma 00UJ we can find an étale ring map $A \to A'$ and a prime ideal $\mathfrak{p}' \subset A'$ lying over $\mathfrak{p}$ such that $\kappa(\mathfrak{p}') = \kappa(\mathfrak{p})$ and such that

$$B' = B \otimes_A A' = C \times D$$

with $A' \to C$ finite and such that the unique prime $\mathfrak{q}'$ of $B \otimes_A A'$ lying over $\mathfrak{q}$ and $\mathfrak{p}'$ corresponds to a prime of C. By Lemma 0BT3 and Algebra, Lemma 00MQ it suffices to show $\omega_{B'/A',\mathfrak{q}'}$ is flat over $A'_{\mathfrak{p}'}$. Since $\omega_{B'/A'} = \omega_{C/A'} \times \omega_{D/A'}$ by Lemma 0BT5 this reduces us to the case where B is finite flat over A. In this case B is finite locally free as an A-module and $\omega_{B/A} = \mathrm{Hom}_A(B, A)$ is the dual finite locally free A-module. $\square$